

# DANO AIRLINES PASSENGER SATISFACTION & SERVICE QUALITY ANALYSIS

## Executive Summary

Customer satisfaction is one of the most important performance indicators in the aviation industry. Satisfied passengers are more likely to remain loyal, recommend the airline to others, and contribute to long-term profitability.

This project analyzed over 130,000 passenger records from Dano Airlines to understand the factors influencing customer satisfaction and identify opportunities for service improvement. The analysis explored passenger demographics, travel type, service quality ratings, flight delays, travel class, and overall satisfaction outcomes.

The findings revealed that overall passenger satisfaction remains relatively low at 43.45%, highlighting significant opportunities for service enhancement. Business Class passengers and business travelers consistently reported higher satisfaction levels, while delays and lower service quality ratings were associated with reduced customer satisfaction.

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## Business Problem

The airline industry operates in an increasingly competitive environment where customer experience plays a critical role in customer retention and revenue generation.

Dano Airlines seeks to understand:

- What factors drive passenger satisfaction?
- How do delays affect customer perception?
- Which customer segments are most satisfied?
- Which services have the strongest impact on customer experience?
- What operational improvements can increase satisfaction rates?

The goal of this project is to provide data-driven recommendations that can help improve passenger experience and strengthen customer loyalty.

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# Dataset Overview

## Dataset Size

- Total Records: 130,000+
- Total Columns: 24

## Key Variables

### Passenger Information:

- Gender
- Age
- Customer Type

### Travel Information:

- Type of Travel
- Travel Class
- Flight Distance

### Operational Metrics:

- Departure Delay
- Arrival Delay

### Service Quality Metrics:

- Online Boarding
- Ease of Online Booking
- Check-in Service
- Seat Comfort
- Leg Room Service
- Cleanliness
- Food and Drink
- In-flight Service
- In-flight Entertainment
- Wi-Fi Service

### Target Variable:

- Satisfaction
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## Data Cleaning & Preparation

The dataset underwent several preprocessing steps to ensure accuracy and consistency.

### Data Cleaning Activities

- Checked for duplicate records
- Handled missing values
- Standardized data formats
- Reviewed categorical variables
- Verified numerical columns

### Feature Engineering

Additional columns were created to improve analysis:

#### *Age Category*

Passengers were grouped into:

- Young Adults
- Adults
- Middle-Aged Adults
- Older Adults
- Seniors

#### *Delay Category*

Flights were categorized as:

- On Time
- Minor Delay
- Moderate Delay
- Severe Delay
- Unknown

#### *Distance Category*

Flight distances were grouped into:

- Short Haul

- Medium Haul
- Long Haul

These new fields enabled deeper segmentation and business analysis.

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## Dashboard KPIs

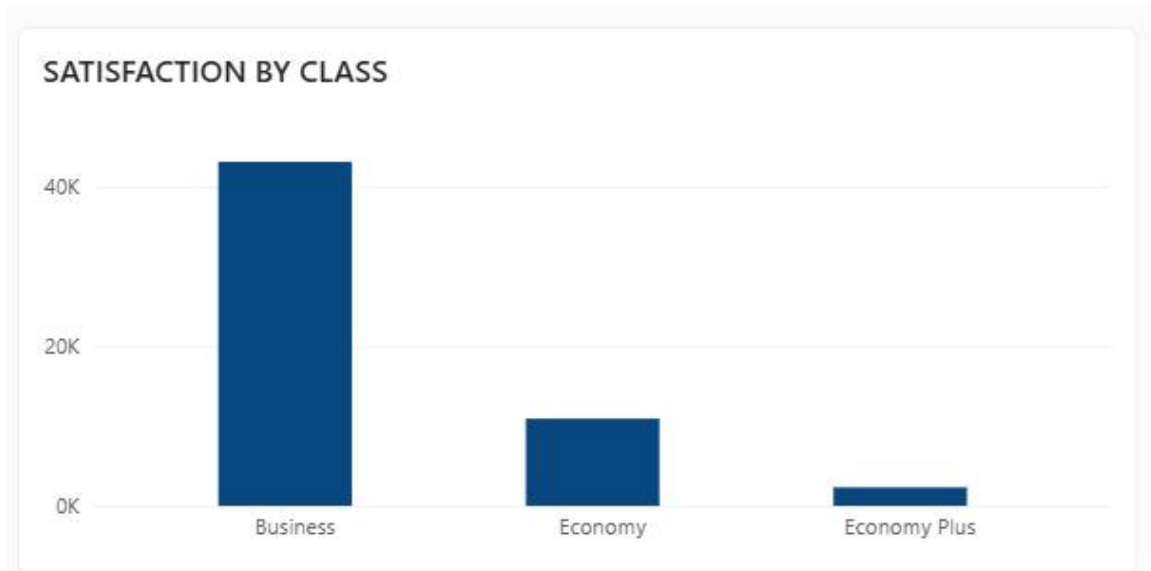
| Total Passengerrrs | Satisfaction Rate | Avg Flight Distance | Average Arrival Delay | Average Cleanliness |
|--------------------|-------------------|---------------------|-----------------------|---------------------|
| 130K               | 43.45%            | 1.19K               | 15.09                 | 3.29                |

Key Performance Indicators:

- Total Passengers: 130K
  - Satisfaction Rate: 43.45%
  - Average Flight Distance: 1.19K Miles
  - Average Arrival Delay: 15.09 Minutes
  - Average Cleanliness Rating: 3.29
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# Analysis & Findings

## Satisfaction by Travel Class



### Findings

Business Class passengers recorded significantly higher satisfaction levels than Economy and Economy Plus passengers.

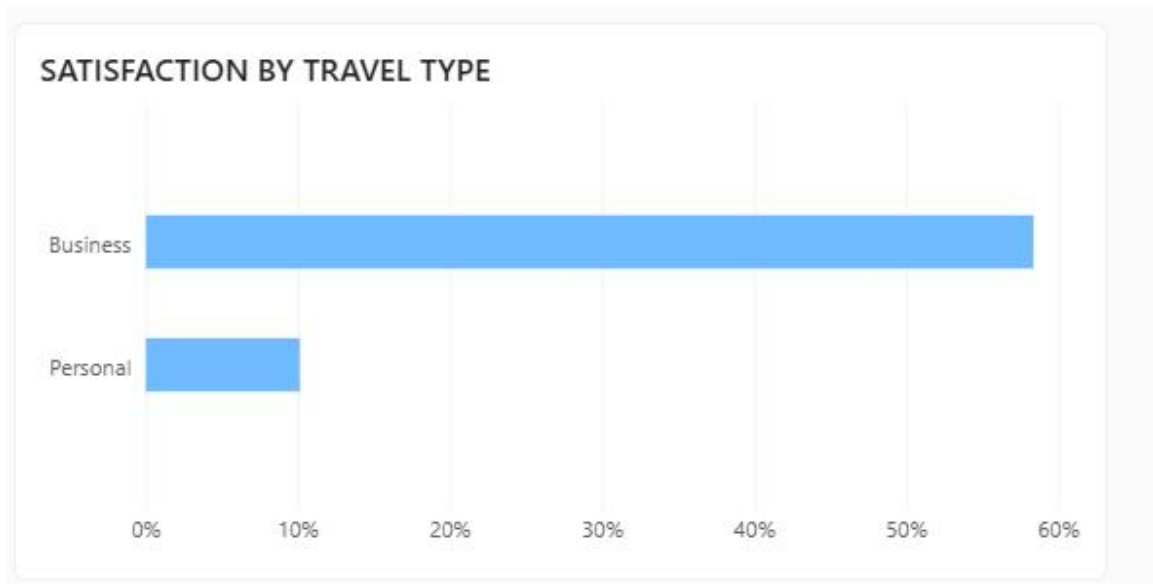
This suggests that premium service offerings have a strong positive effect on customer perception.

### Business Implication

Passengers are willing to reward better service quality with higher satisfaction ratings.

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## Satisfaction by Travel Type



### Findings

Business travelers reported substantially higher satisfaction than personal travelers.

Possible reasons include:

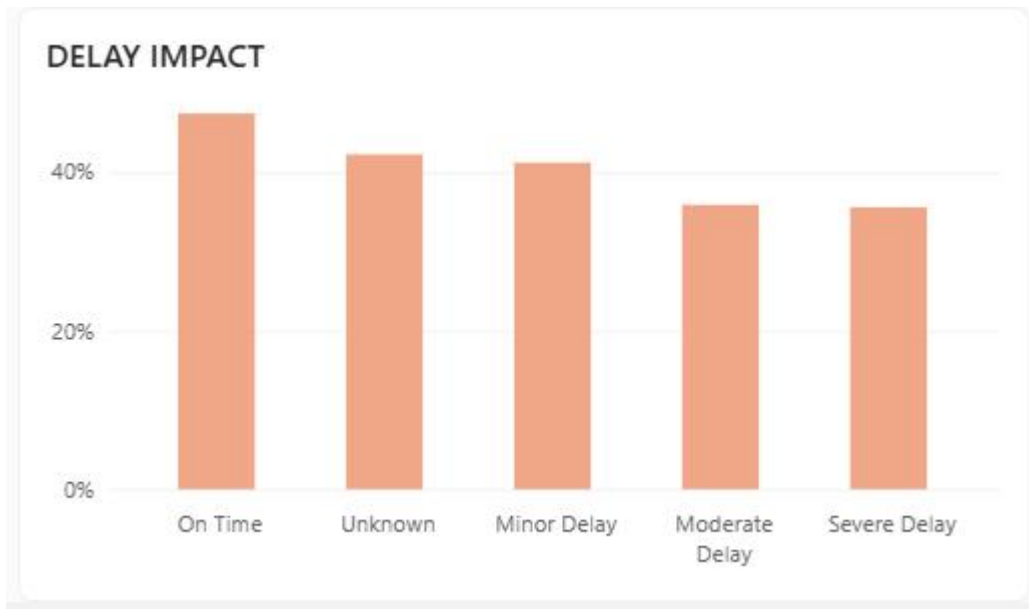
- Priority services
- Better flight schedules
- Greater use of premium cabins
- More frequent flyer benefits

### Business Implication

Personal travelers represent a significant opportunity for customer experience improvement.

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## Delay Impact Analysis



### Findings

Passengers on on-time flights reported the highest satisfaction rates.

As delays increased, customer satisfaction consistently declined.

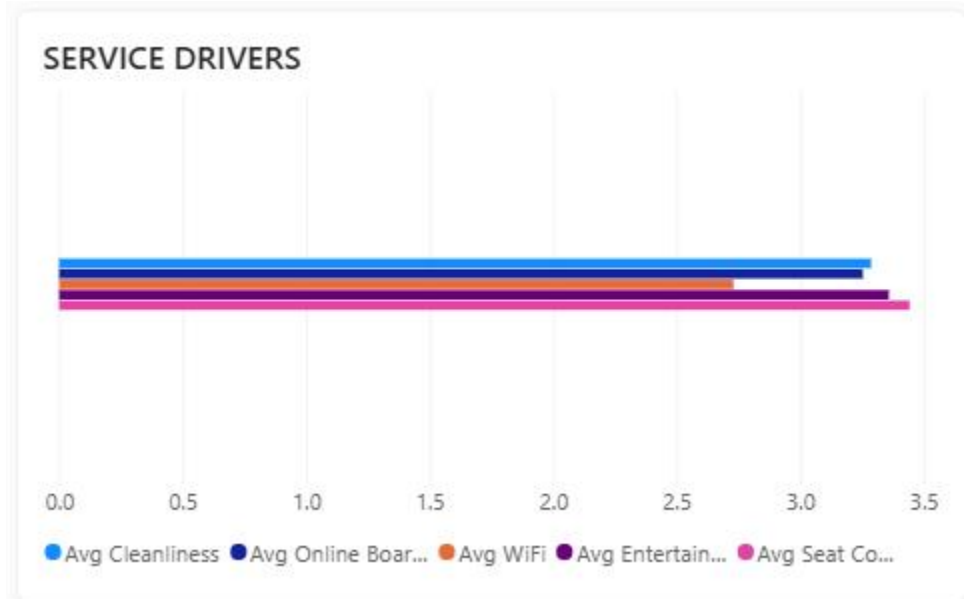
Moderate and severe delays were associated with the lowest satisfaction outcomes.

### Business Implication

Operational efficiency remains one of the strongest drivers of customer experience.

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## Service Drivers Analysis



### Findings

The strongest service drivers included:

- Seat Comfort
- Online Boarding
- Cleanliness
- In-flight Entertainment
- Wi-Fi Service

These services consistently received higher ratings and demonstrated strong relationships with overall satisfaction.

### Business Implication

Investments in service quality directly contribute to improved customer experience.

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# Key Insights

## Insight 1

Overall satisfaction remains below 50%, indicating substantial room for improvement.

## Insight 2

Business Class significantly outperforms Economy Class in customer satisfaction.

## Insight 3

Business travelers are generally more satisfied than leisure travelers.

## Insight 4

Flight delays have a measurable negative impact on passenger experience.

## Insight 5

Comfort, cleanliness, and digital services strongly influence satisfaction outcomes.

## Insight 6

Improving operational reliability may have a greater impact on satisfaction than introducing new services.

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# Recommendations

## Operational Recommendations

### 1. Reduce Flight Delays

Actions:

- Improve scheduling efficiency
- Optimize turnaround processes
- Strengthen operational monitoring

Expected Outcome:

Higher passenger satisfaction and improved reliability perception.

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## 2. Improve Economy Class Experience

Actions:

- Upgrade seating comfort
- Enhance cabin cleanliness
- Improve food quality

Expected Outcome:

Reduction in the satisfaction gap between Economy and Business Class.

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## 3. Strengthen Digital Passenger Experience

Actions:

- Improve online boarding systems
- Enhance mobile services
- Simplify booking processes

Expected Outcome:

Higher convenience ratings and customer satisfaction.

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## 4. Enhance In-Flight Comfort

Actions:

- Improve seat ergonomics
- Increase legroom where feasible
- Upgrade entertainment offerings

Expected Outcome:

Stronger customer loyalty and repeat bookings.

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## 5. Improve Customer Communication During Delays

Actions:

- Provide real-time updates
- Offer proactive notifications
- Improve support responsiveness

Expected Outcome:

Reduced frustration during service disruptions.

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## 6. Develop Personal Traveler Loyalty Programs

Actions:

- Offer targeted promotions
- Introduce rewards programs
- Improve value-added services

Expected Outcome:

Increased satisfaction among personal travelers.

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## Conclusion

The analysis demonstrates that customer satisfaction at Dano Airlines is heavily influenced by operational performance and service quality. While Business Class and business travelers report positive experiences, there is significant opportunity to improve satisfaction among Economy passengers and personal travelers.

By reducing delays, improving service quality, and investing in customer-focused initiatives, Dano Airlines can significantly enhance customer satisfaction and strengthen its competitive position within the airline industry.

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# Full Dashboard Screenshots

